

# BERYL HOVIS-AFFLERBACH

## ASTRONOMY PHD CANDIDATE

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## EDUCATION

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|--------------|---------------------------------------------------------------------------------------------------------------------------------------|--------------|
| 2023–present | <b>Northwestern University</b><br>PhD candidate in Astronomy; advisor: Prof. Allison Strom<br>Master of Science, Astronomy, June 2025 | Evanston, IL |
| 2018–2023    | <b>California Institute of Technology</b><br>Bachelor of Science, Astrophysics, June 2023                                             | Pasadena, CA |

## AWARDS & FUNDING

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|------|----------------------------------------------------|------------------------------------|
| 2023 | NSF Graduate Research Fellow                       | NSF                                |
| 2022 | Vodopia-Hasson Poster Competition Award            | California Institute of Technology |
| 2022 | George W. Housner Fund Recipient                   | California Institute of Technology |
| 2021 | Chambliss Undergraduate Poster Award, AAS 238      | AAS                                |
| 2021 | Carnegie Observatories Summer Student Poster Award | Carnegie Observatories             |
| 2019 | George W. Housner Fund Recipient                   | California Institute of Technology |

## PUBLICATIONS

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### FIRST AUTHOR

4. *The mass distribution of stars stripped in binaries: The effect of metallicity*  
**Hovis-Afflerbach, B.**, Göteborg, Y., Schootemeijer, A., Klencki, J., Strom, A. L., Ludwig, B. A., Drout, M. R., 2025, *A&A*, 697, A239
3. *Partially Erupted Prominence Material as a Diagnostic of Coronal Mass Ejection Trajectory*  
**Hovis-Afflerbach, B.**, Thompson, B. J., & Mason, E. I., 2023, *SpWea*, 21, e2022SW003256
2. *Two New Methods for Counting and Tracking the Evolution of Polar Faculae*  
**Hovis-Afflerbach, B.** & Pesnell, W. D., 2022, *SoPh*, 297, 48
1. *Identifying and Repairing Catastrophic Errors in Galaxy Properties Using Dimensionality Reduction*  
**Hovis-Afflerbach, B.**, Steinhardt, C. L., Masters, D., & Salvato, M., 2021, *ApJ*, 908, 148

### CO-AUTHOR

1. *The BUFFALO HST Survey*  
Steinhardt, C. L., et al., incl. **Hovis-Afflerbach, B.**, 2020, *ApJS*, 247, 64

## SELECTED ACADEMIC PRESENTATIONS

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### TALKS

- 2026 May **Binary Products as Ionizing Sources: Testing & Comparing Models Across Cosmic Time** Baltimore, MD  
The Metal-Poor Frontier: Understanding Low-Metallicity Stars and Galaxies Across Cosmic Time, STScI
- 2025 Sep **Finding He II: Testing Novel Models of Binary Populations Across Cosmic Time** Ensenada, Mexico  
IAUS 402: Massive stars across redshifts in the era of JWST and large surveys
- 2024 Feb **Missing Massive Stripped Stars at Low Metallicity** Evanston, IL  
CIERA Theory Group Meeting, Northwestern University

### POSTERS

- 2025 May **Testing Novel Models of Binary Populations Across Cosmic Time** Austin, TX  
Inaugural Cosmic Frontier Center Conference, UT Austin
- 2024 May **Missing Massive Stripped Stars at Low Metallicity: Implications for High-Redshift Galaxies** New York, NY  
First Stars VII, Flatiron Institute
- 2023 Jan **A New Way to Model the Pre-Supernova Evolution of 8–11  $M_{\odot}$  Stars** Seattle, WA  
American Astronomical Society, AAS Meeting #241
- 2022 Jun **Testing Late Stellar Expansion with Populations of Stars Stripped in Binaries** Pasadena, CA  
American Astronomical Society, AAS Meeting #240
- 2021 Jun **Two New Methods for Counting and Tracking the Evolution of Polar Faculae** Virtual  
American Astronomical Society, AAS Meeting #238; won Chambliss Award

## MENTORING & TEACHING

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### RESEARCH MENTORING

- 2026 Summer **Research Mentor, REACH Program** Northwestern University  
Supervised Olivia Turner (Glenbrook South High School) on a project investigating how  $\alpha$ -enhancement affects the radius evolution of massive stars; results will be included in an upcoming paper.

### ACADEMIC & PEER MENTORING

- 2025–present **CIERA Mentorship Network** Northwestern University  
Mentor an undergraduate student in astronomy for multiple years.
- 2024–present **Graduate Peer Mentor, Physics & Astronomy** Northwestern University  
Mentor first-year graduate students.
- 2022–present **Undergraduate Mentor, CASSI Summer Program** Carnegie Observatories  
As a program alum, mentor 1–2 students each summer.

### TEACHING

- 2025 Spring **ASTRON 220-1: Intro to Astrophysics I** Northwestern University  
Teaching assistant for a course on the origin and evolution of star and planet systems, focusing on the underlying physical processes and observational techniques.
- 2025 Winter **PHYSICS 135-3: General Physics** Northwestern University  
Teaching assistant for introductory waves & optics.
- 2024 Fall **PHYSICS 135-2: General Physics** Northwestern University  
Teaching assistant for introductory electricity & magnetism.

## LEADERSHIP & SERVICE

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| 2026–present | <b>Referee, The Astrophysical Journal</b>                                                                                                                                                                                        | AAS                                |
| 2025–present | <b>CIERA Stars Meeting</b><br>Founded and organize a biweekly meeting where researchers who study stars from different perspectives share research, discuss recent papers and results, and find opportunities for collaboration. | Northwestern University            |
| 2025–present | <b>President, Physics &amp; Astronomy Graduate Student Council</b>                                                                                                                                                               | Northwestern University            |
| 2025–2026    | <b>Science Editor, CIERA 15th Anniversary Coffee Table Book</b>                                                                                                                                                                  | Northwestern University            |
| 2024–2025    | <b>Secretary, Physics &amp; Astronomy Graduate Student Council</b>                                                                                                                                                               | Northwestern University            |
| 2019–2022    | <b>Equity &amp; Title IX Advocate</b><br>Supported peers in cases involving Equity and Title IX violations; organized meetings to welcome new students and share Title IX resources.                                             | California Institute of Technology |
| 2019–2022    | <b>Board of Control Representative</b><br>Heard cases of alleged academic Honor Code violations (over 100 hours).                                                                                                                | California Institute of Technology |

## OUTREACH

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|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| 2024–present              | <b>Research Experiences in Astronomy at CIERA for High School Students (REACH)</b><br>Developed and present an introductory lesson on stars with an interactive component; serve on panels and tutor students in Python. | Northwestern University |
| 2025<br>Volunteer         | <b>CIERA Reaches Out to Chicago Schools (CROCS)</b><br>Visited local high schools to give presentations and demonstrations on spectroscopy, solar observations, and gravitational waves.                                 | Northwestern University |
| 2024–present<br>Volunteer | <b>Astronomy Conversations</b><br>Lead discussions with visitors and answer questions about astronomy in the Adler Planetarium's Space Visualization Lab.                                                                | Adler Planetarium       |
| 2023–present<br>Volunteer | <b>Astronomy on Tap</b><br>Help organize events for the Chicago chapter; gave a public talk on binary stars in February 2025.                                                                                            | Northwestern University |

## SOFTWARE

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Python • MESA • POSYDON • Cloudy • Pypelt • Linux • IDL